

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



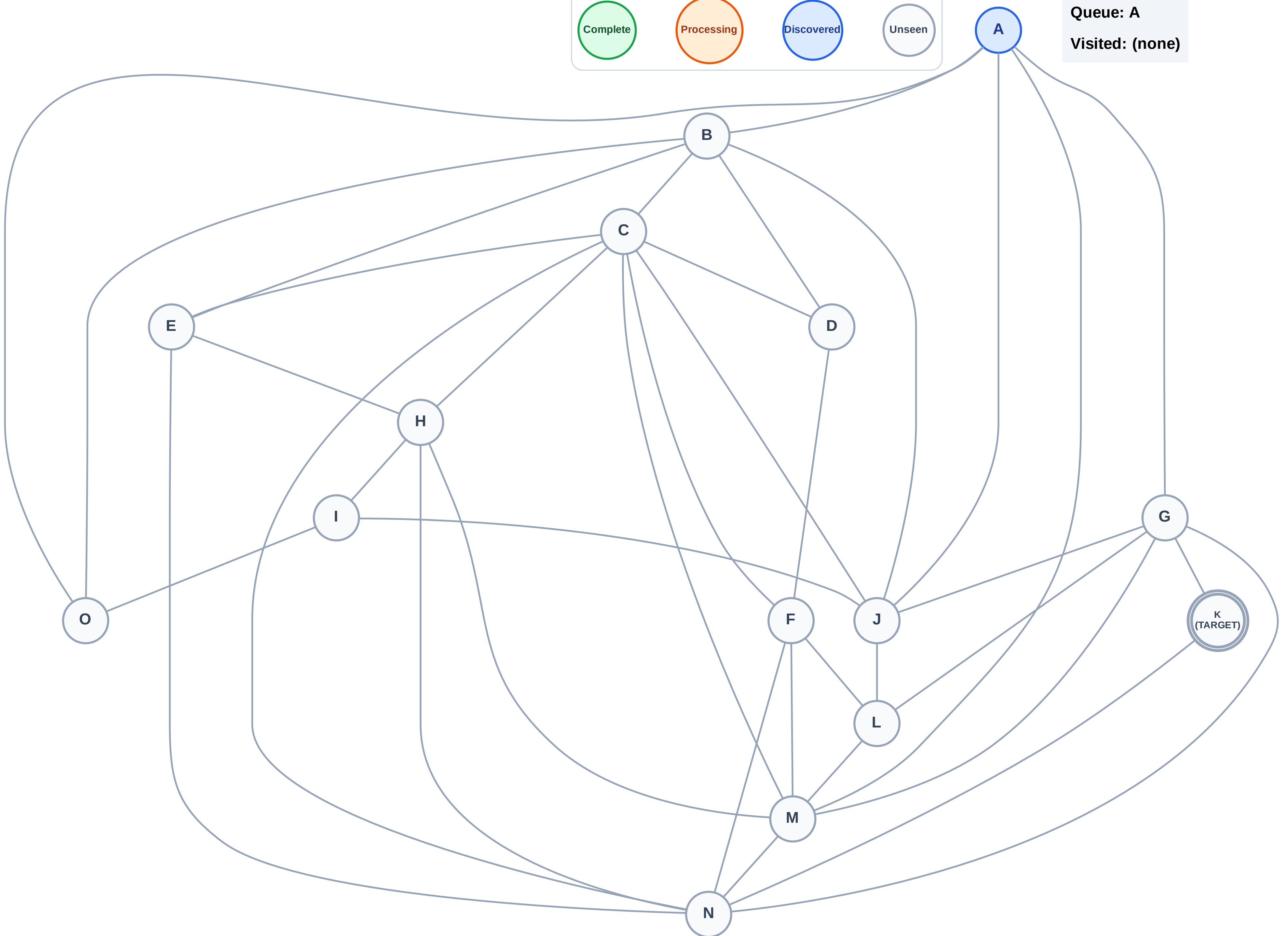
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

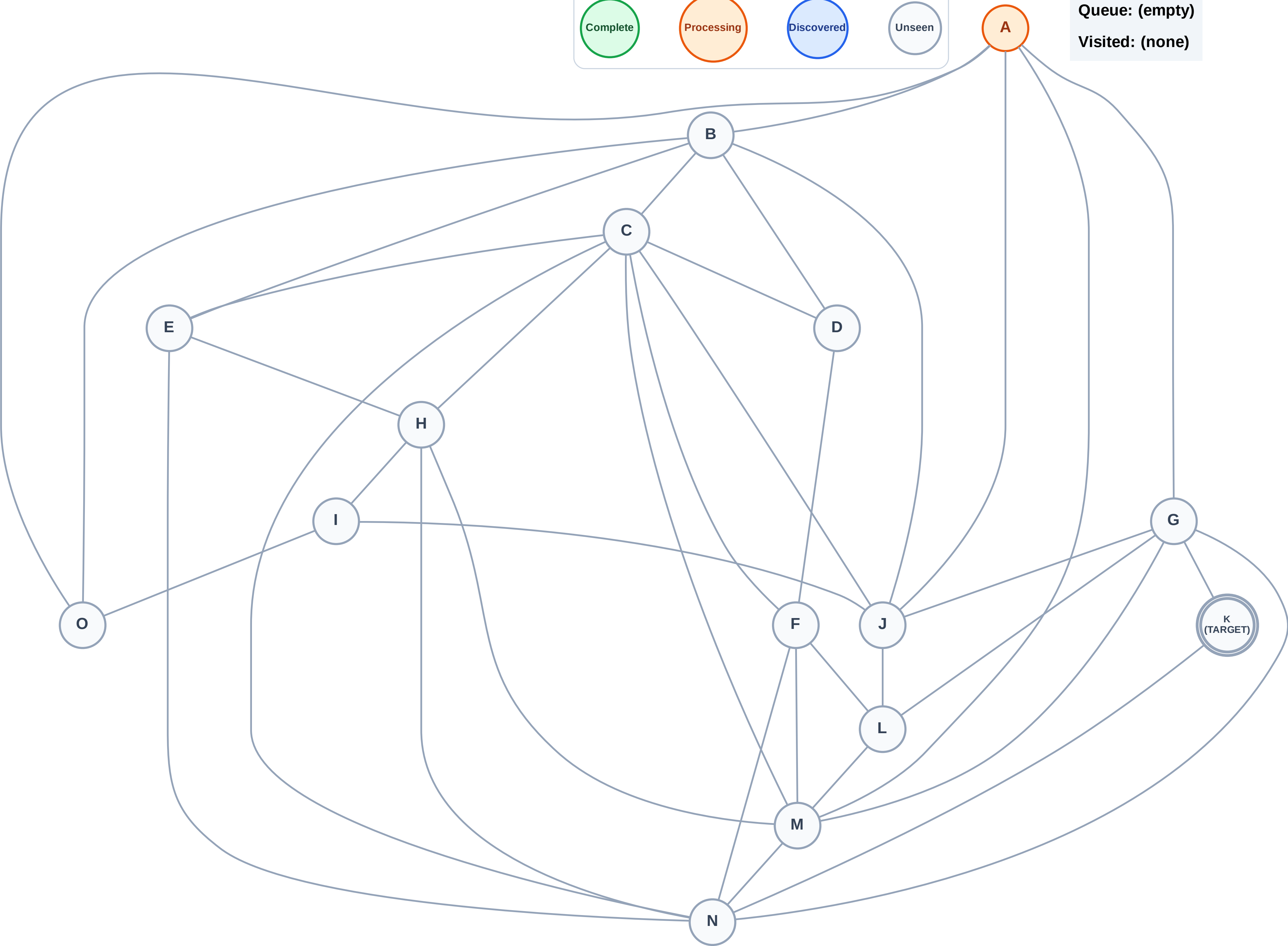
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, G, J, M, O from A

Legend



Complete



Processing



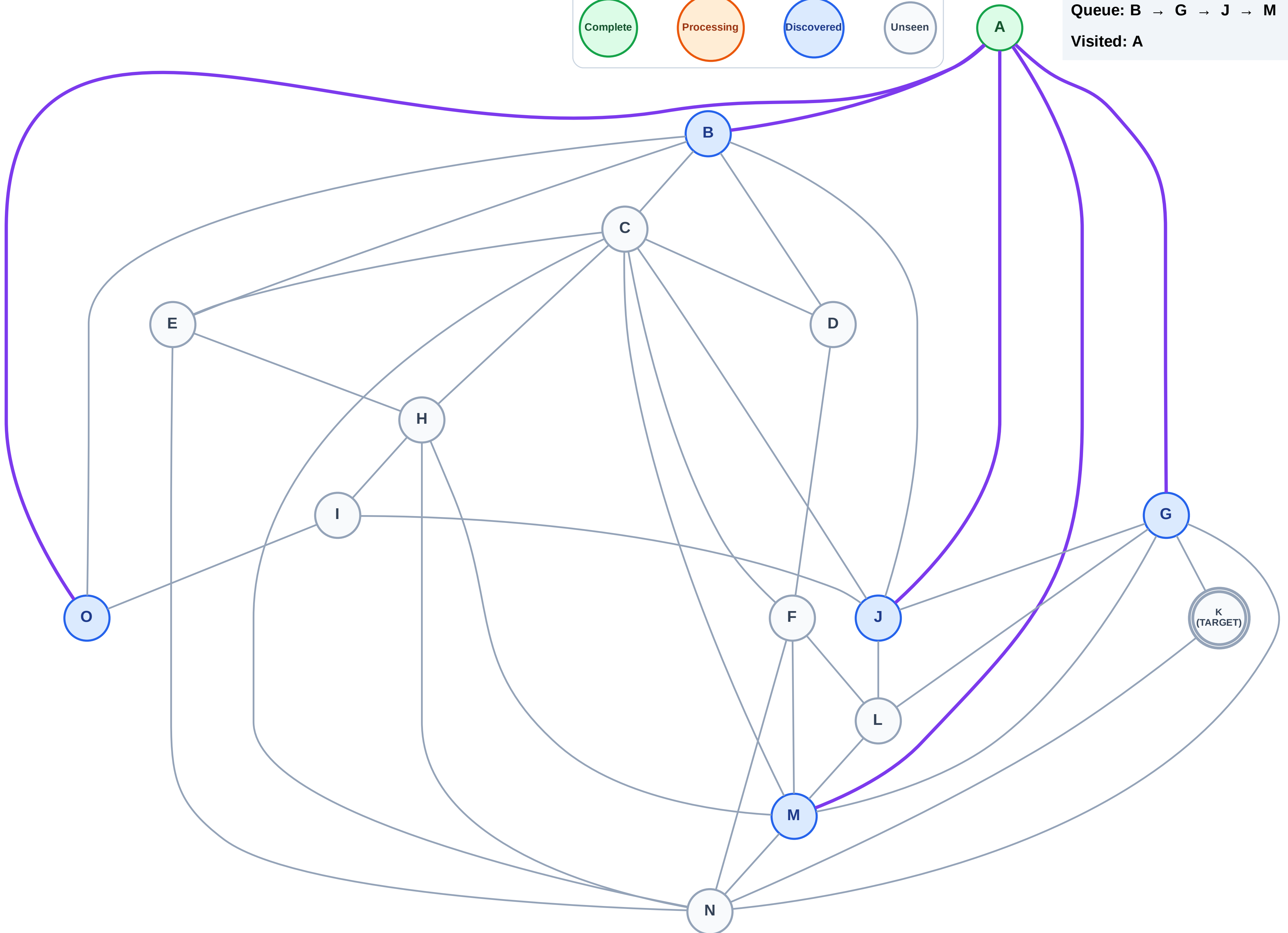
Discovered



Unseen

Queue: B → G → J → M → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

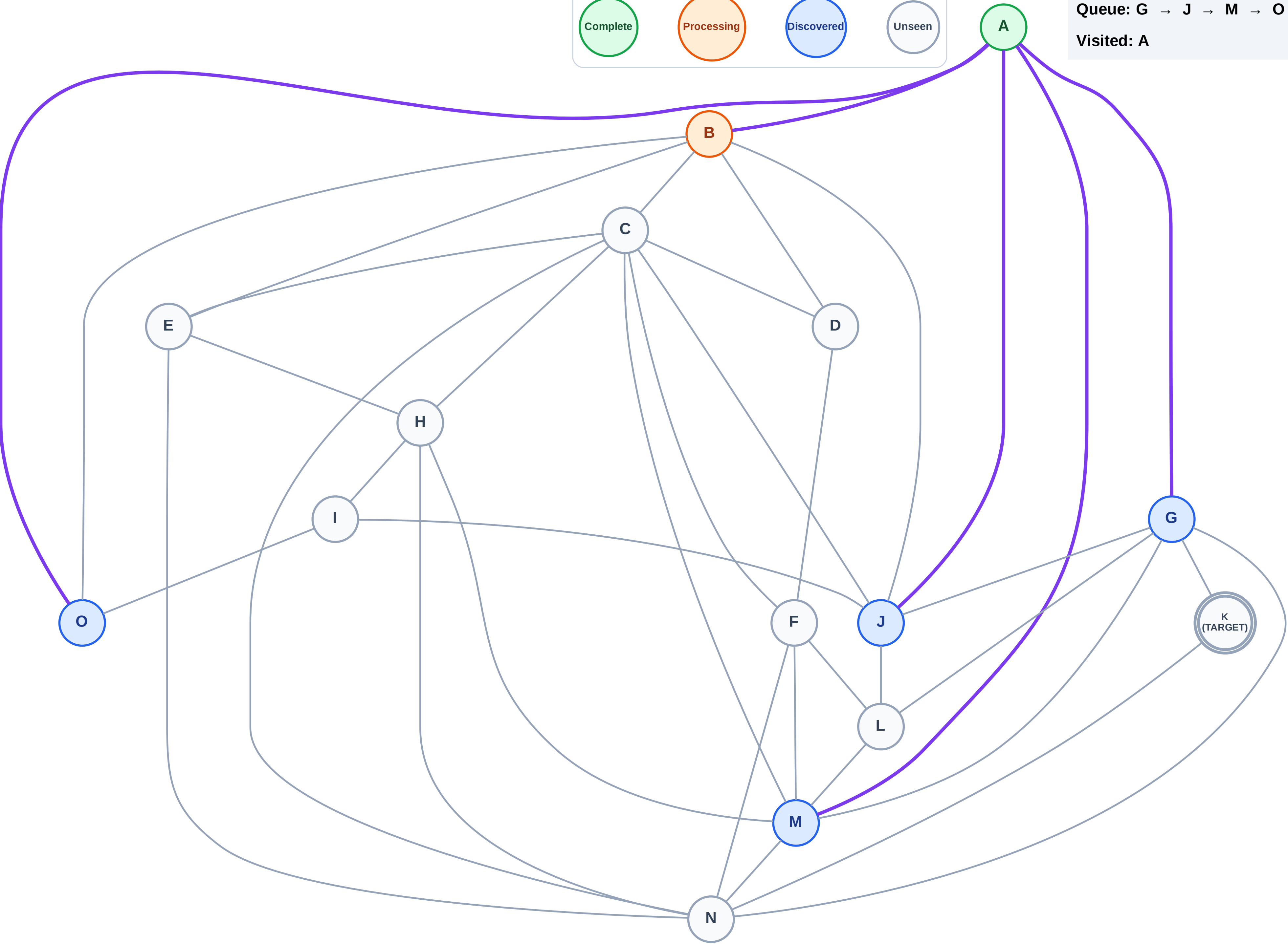
Complete

Processing

Discovered

Unseen

Queue: G → J → M → O
Visited: A



BFS Traversal

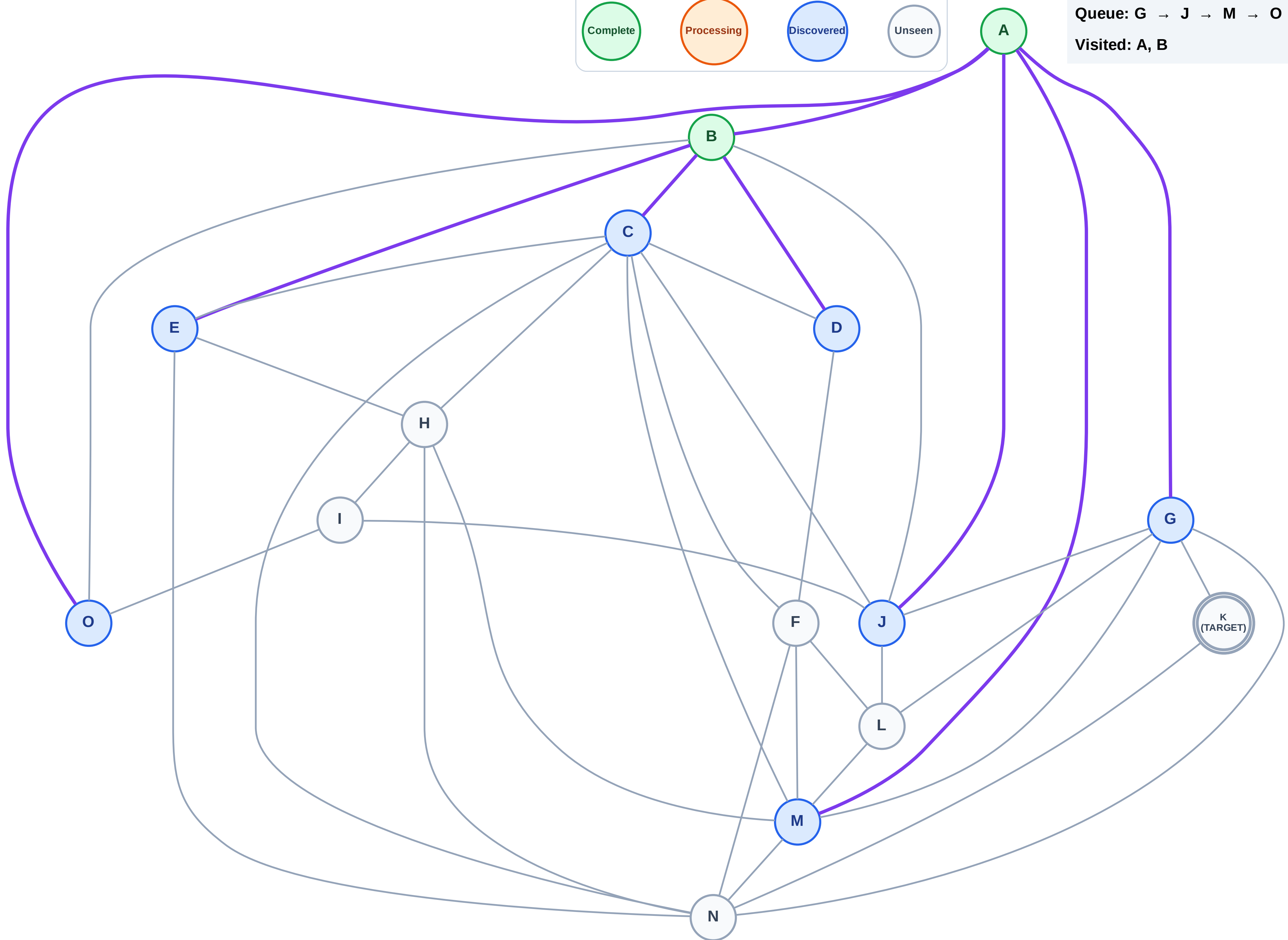
Step 5: Discover C, D, E from B

Legend



Queue: G → J → M → O → C → D → E

Visited: A, B



BFS Traversal

Step 6: Process node G

Legend



Complete



Processing



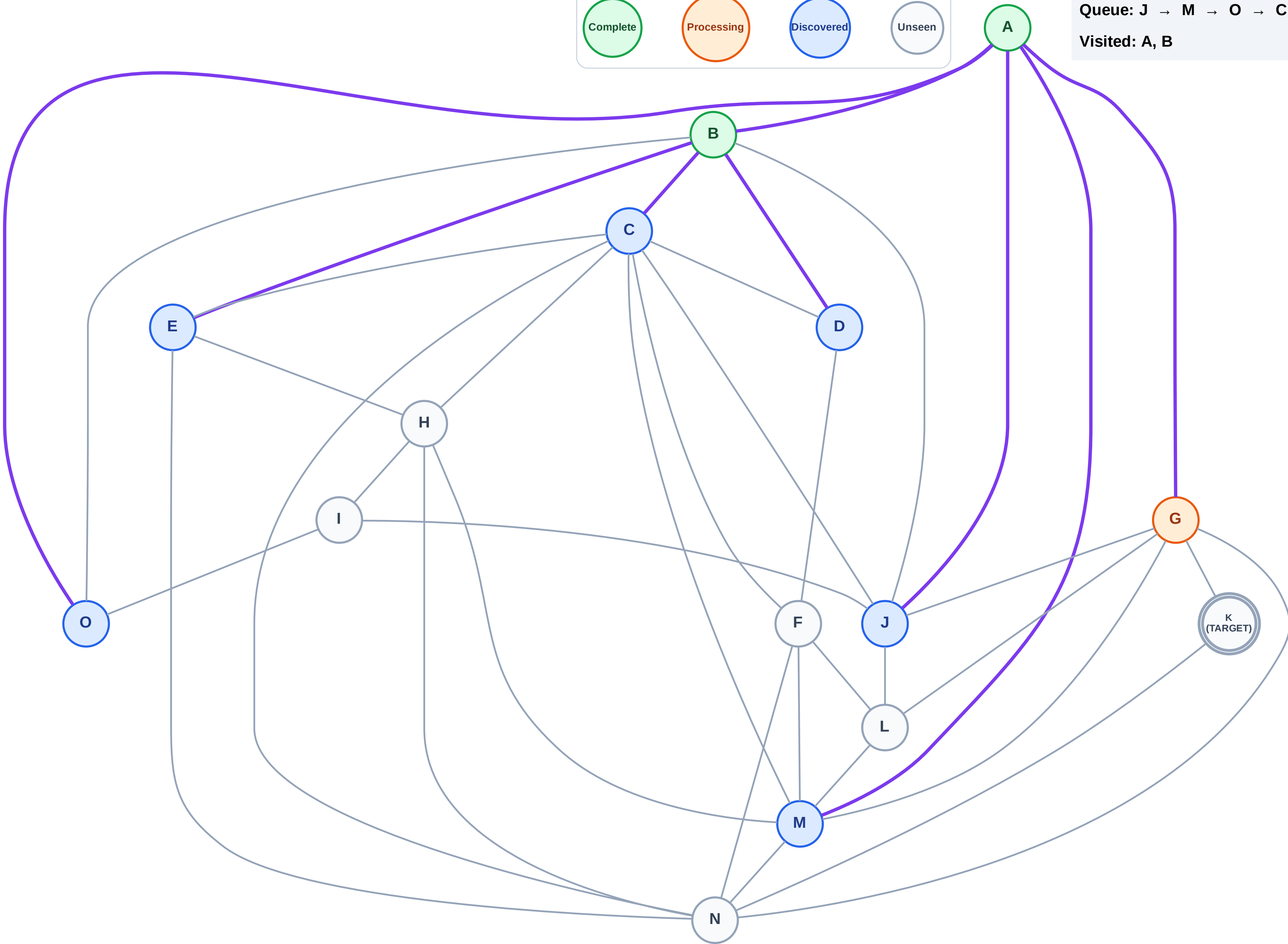
Discovered



Unseen

Queue: J → M → O → C → D → E

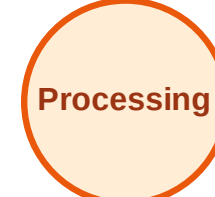
Visited: A, B



BFS Traversal

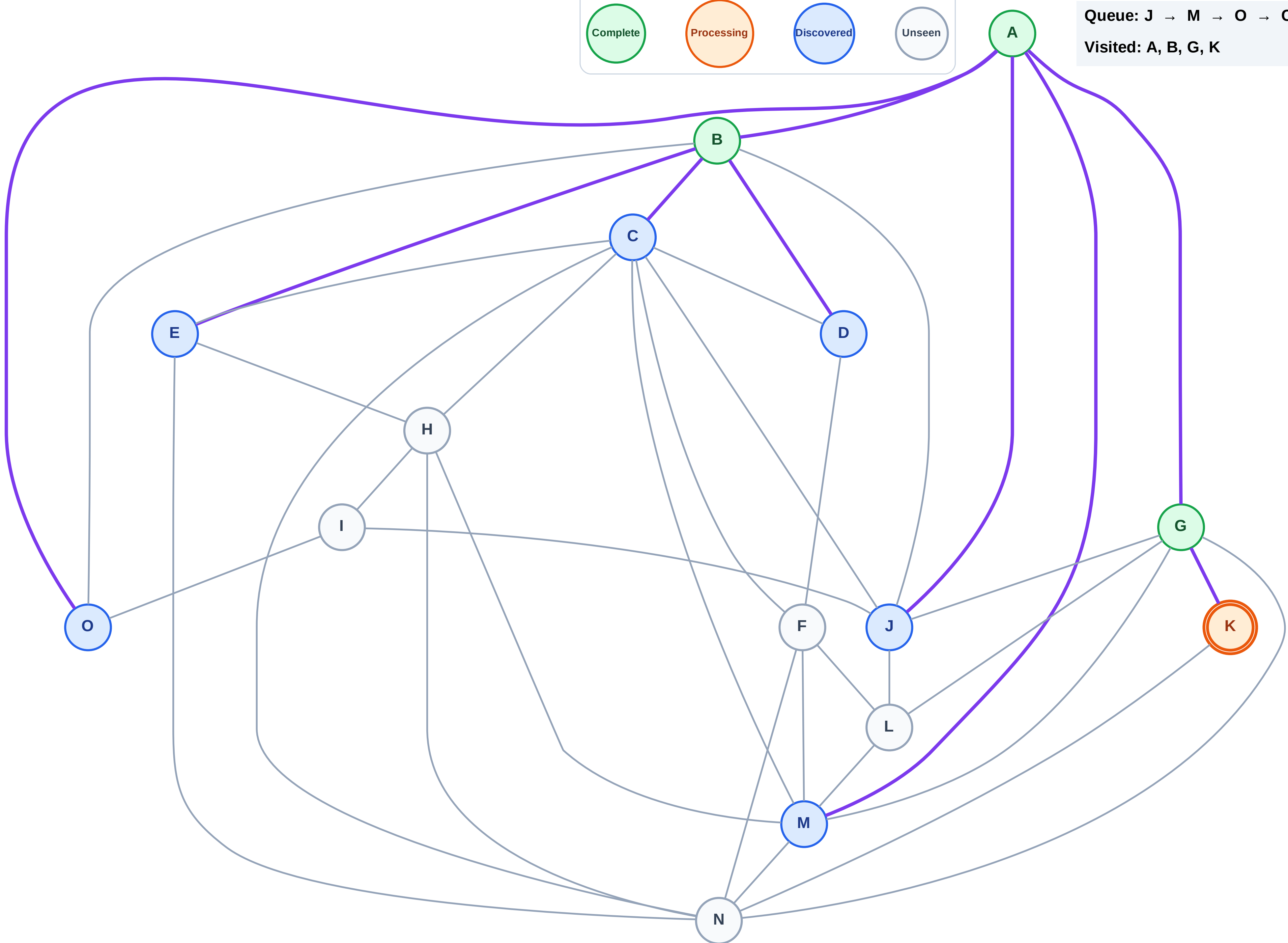
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → G → K

Legend



Queue: J → M → O → C → D → E

Visited: A, B, G, K



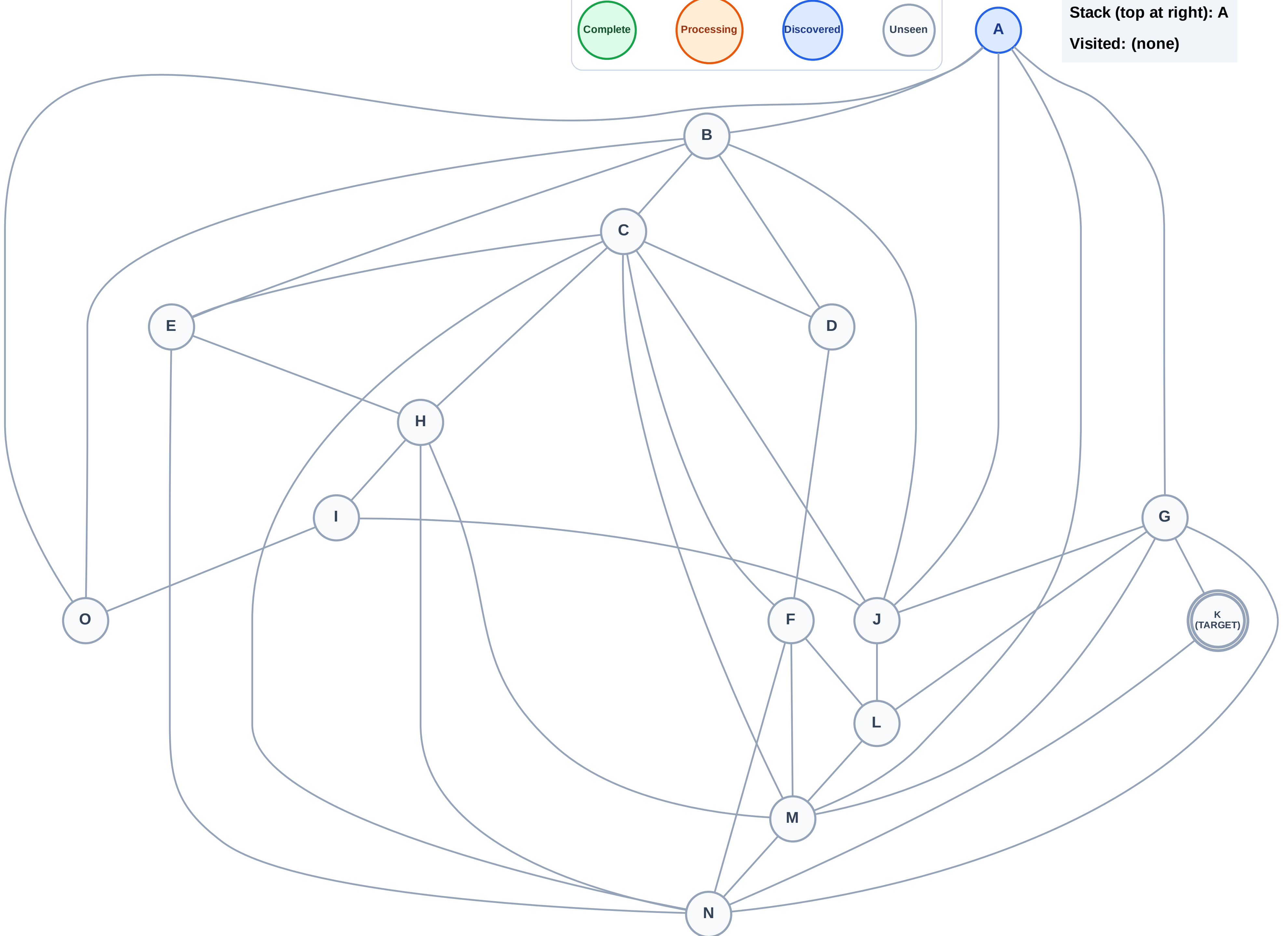
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

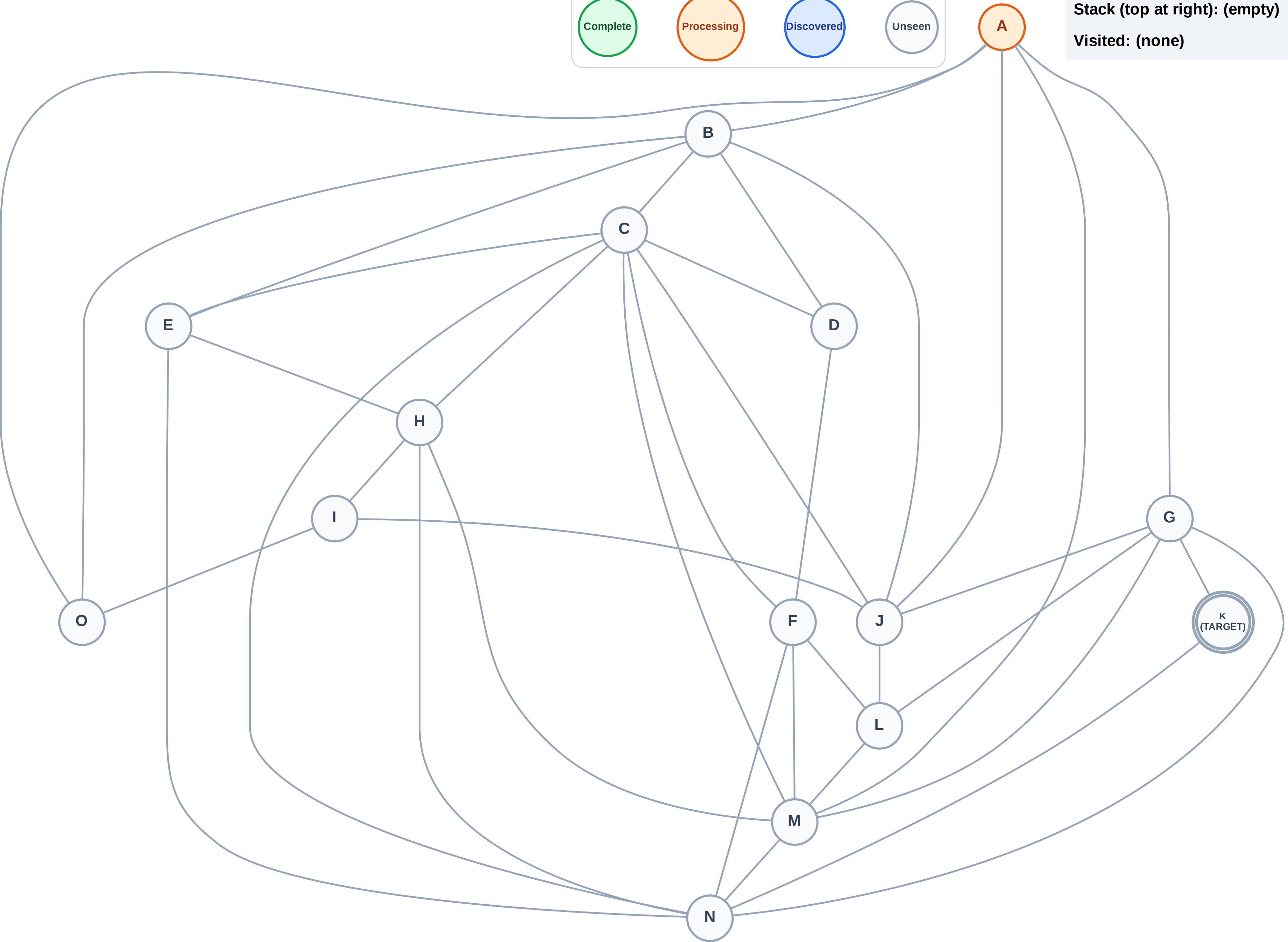
Complete

Processing

Discovered

Unseen

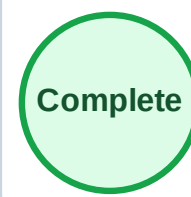
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover O, M, J, G, B from A

Legend



Complete



Processing



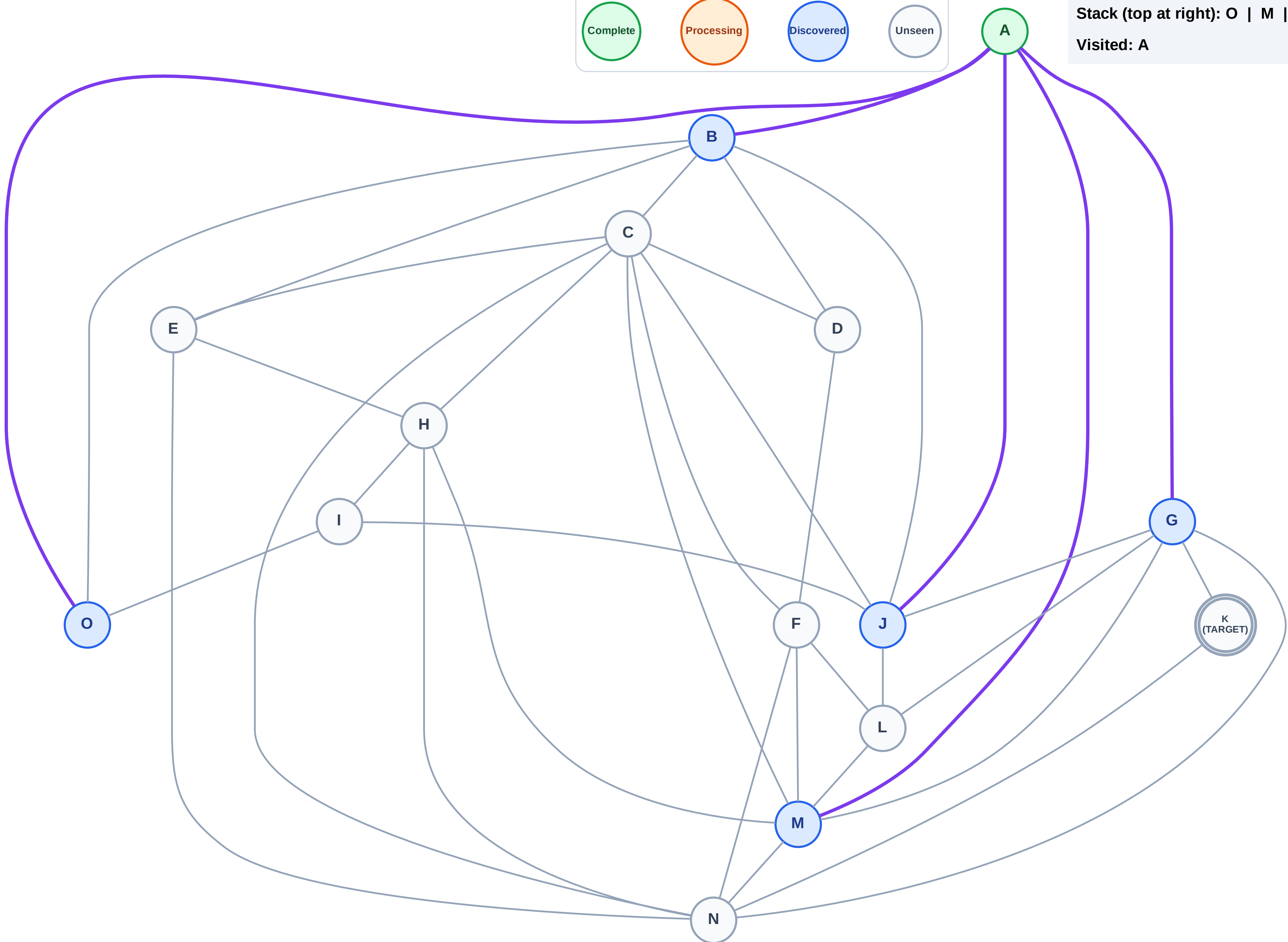
Discovered



Unseen

Stack (top at right): O | M | J | G | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



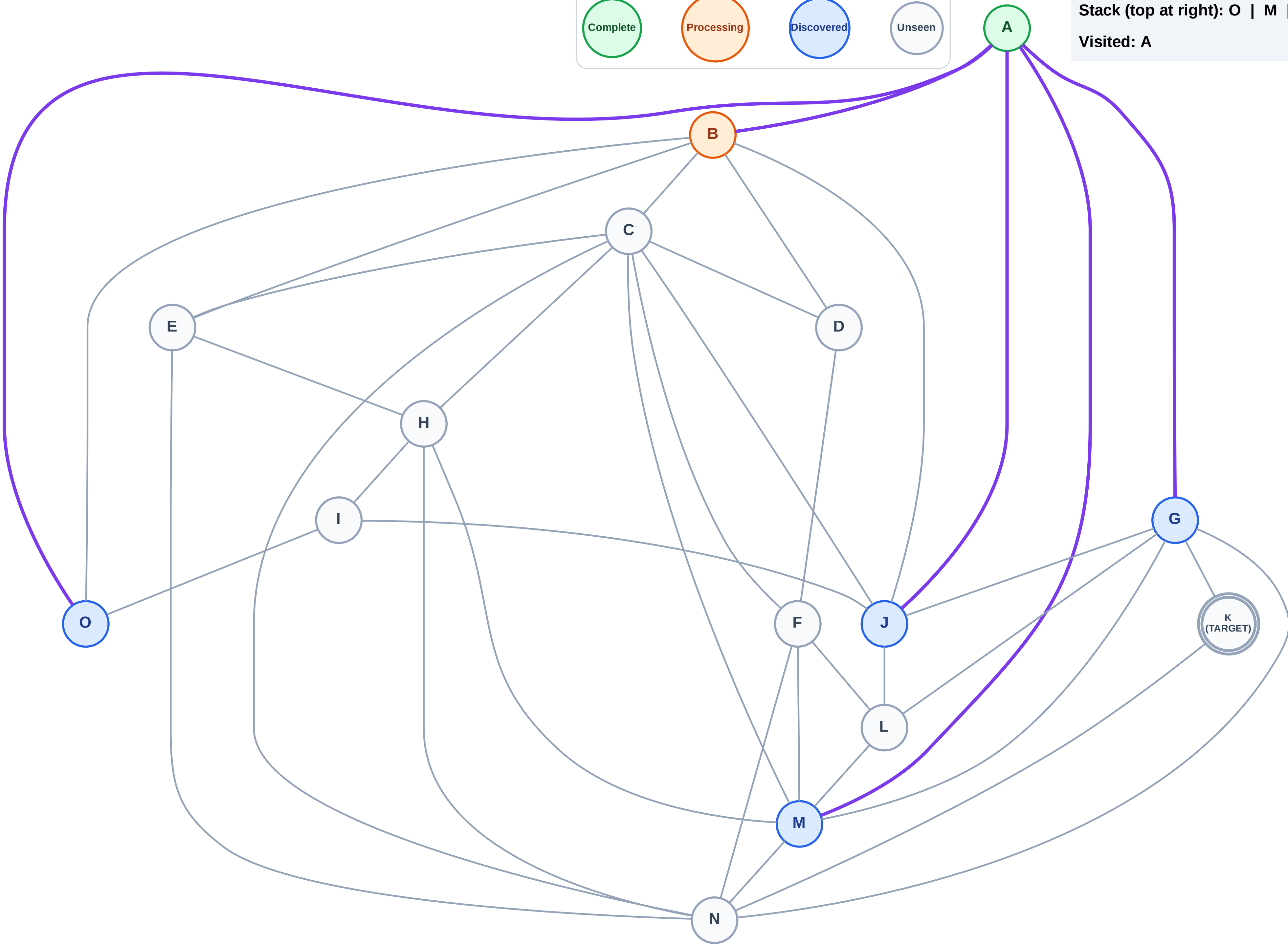
Discovered



Unseen

Stack (top at right): O | M | J | G

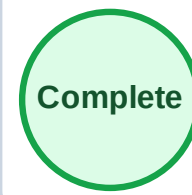
Visited: A



DFS Traversal

Step 5: Discover E, D, C from B

Legend



Complete



Processing



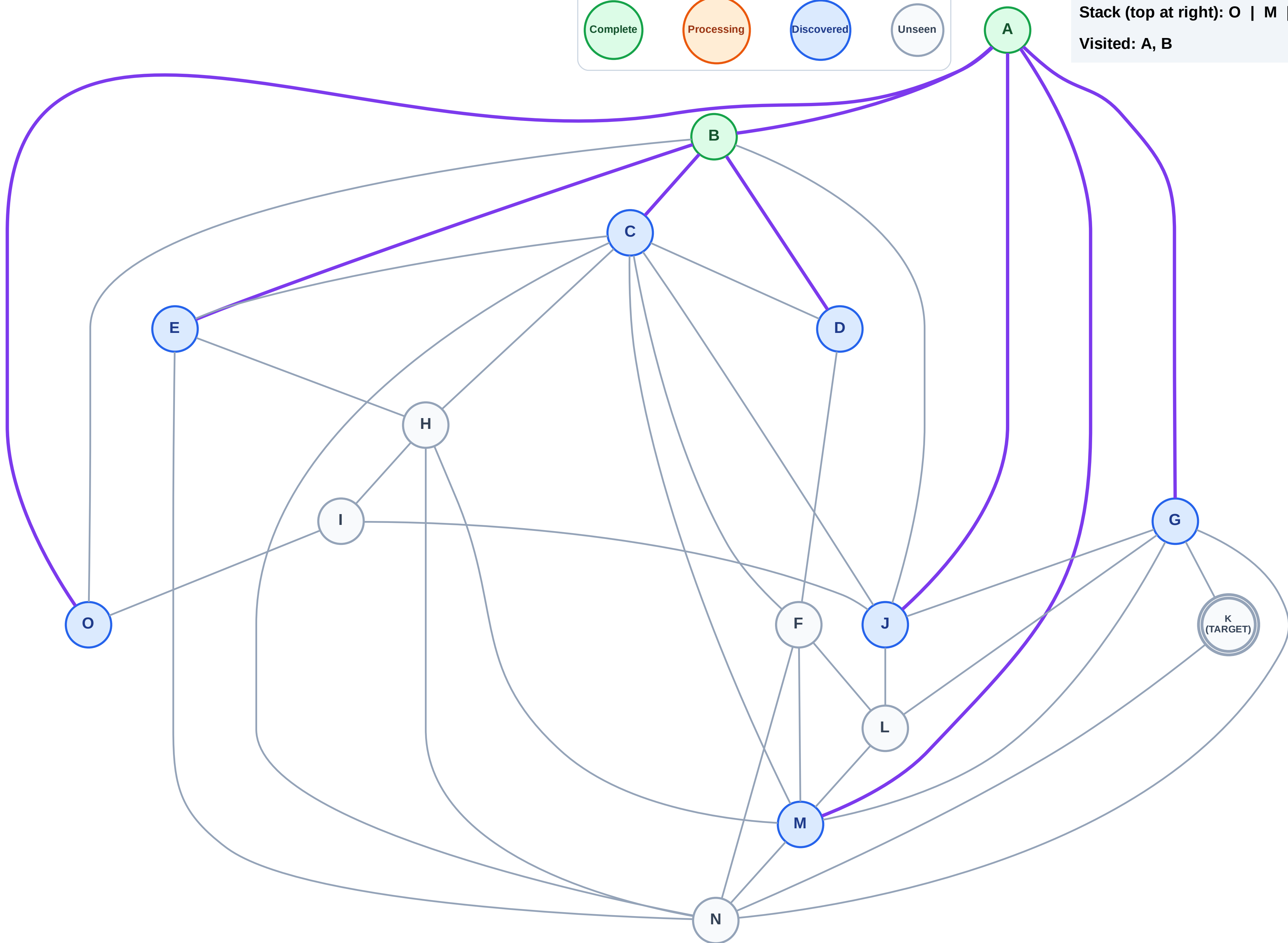
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | C

Visited: A, B

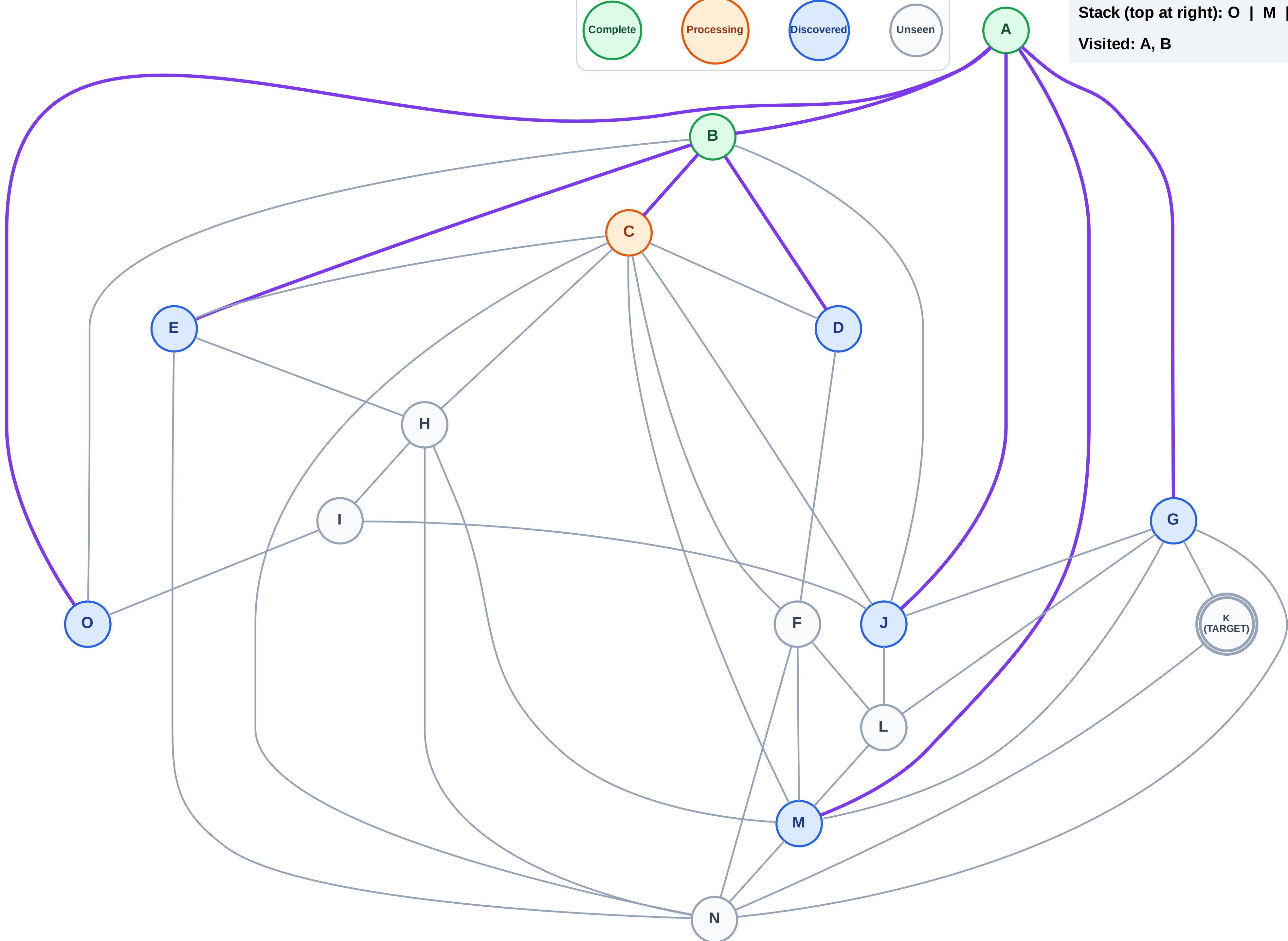


Step 6: Process node C

Legend



Visited: A B



DFS Traversal

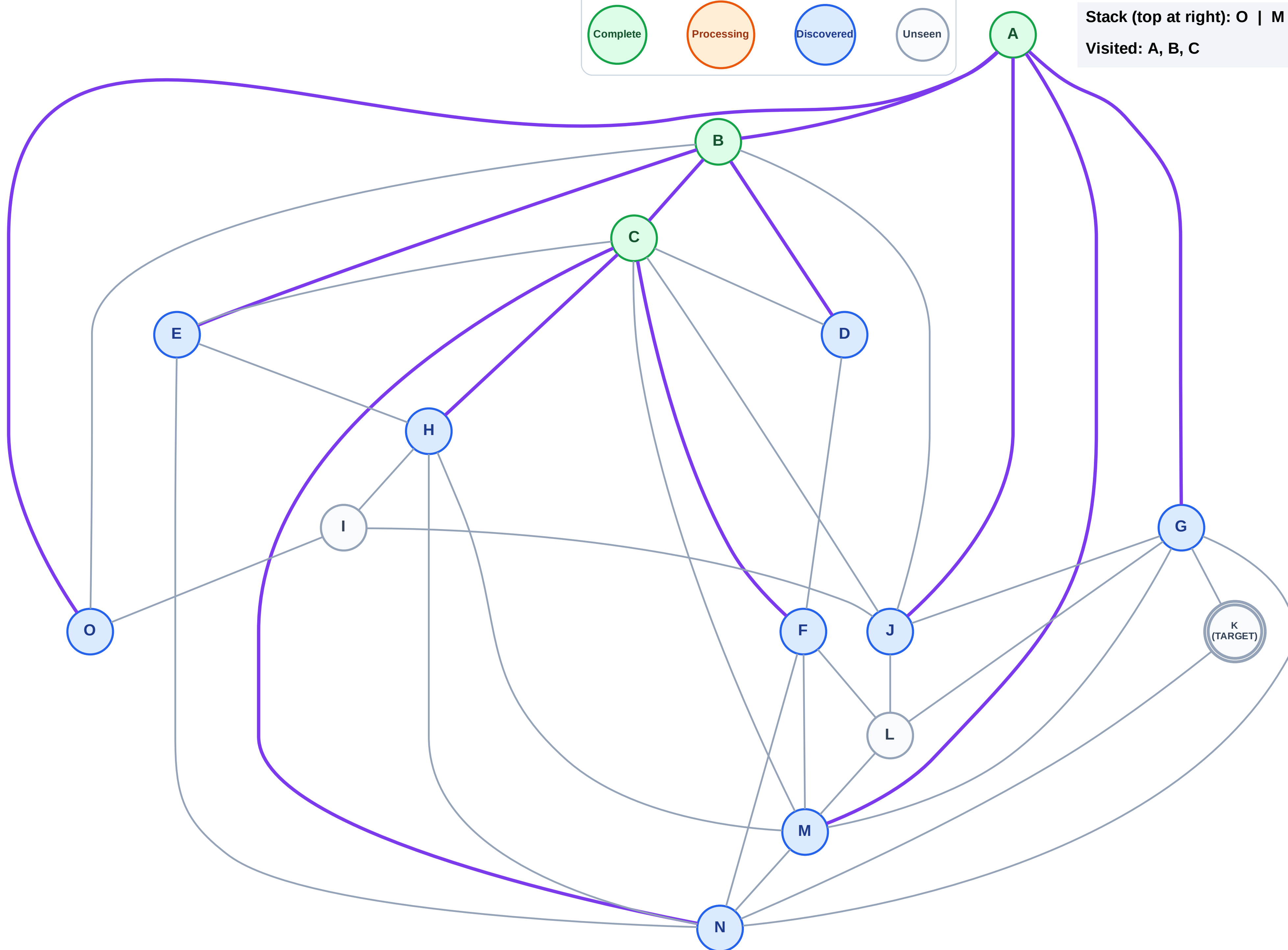
Step 7: Discover N, H, F from C

Legend



Stack (top at right): O | M | J | G | E | D | N | H | F

Visited: A, B, C



DFS Traversal

Step 8: Process node F

Legend

Complete

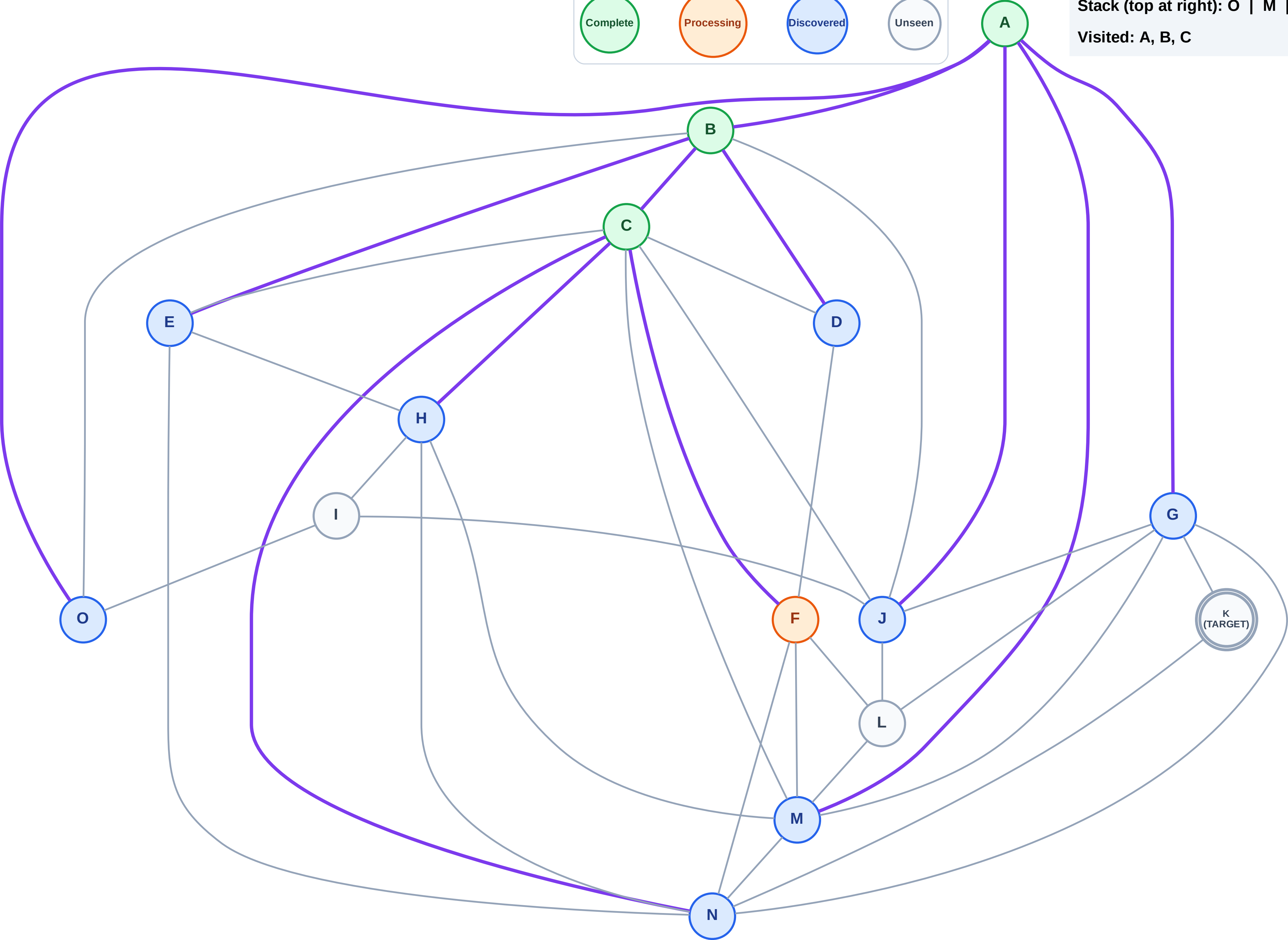
Processing

Discovered

Unseen

Stack (top at right): O | M | J | G | E | D | N | H

Visited: A, B, C



DFS Traversal

Step 9: Discover L from F

Legend

Complete

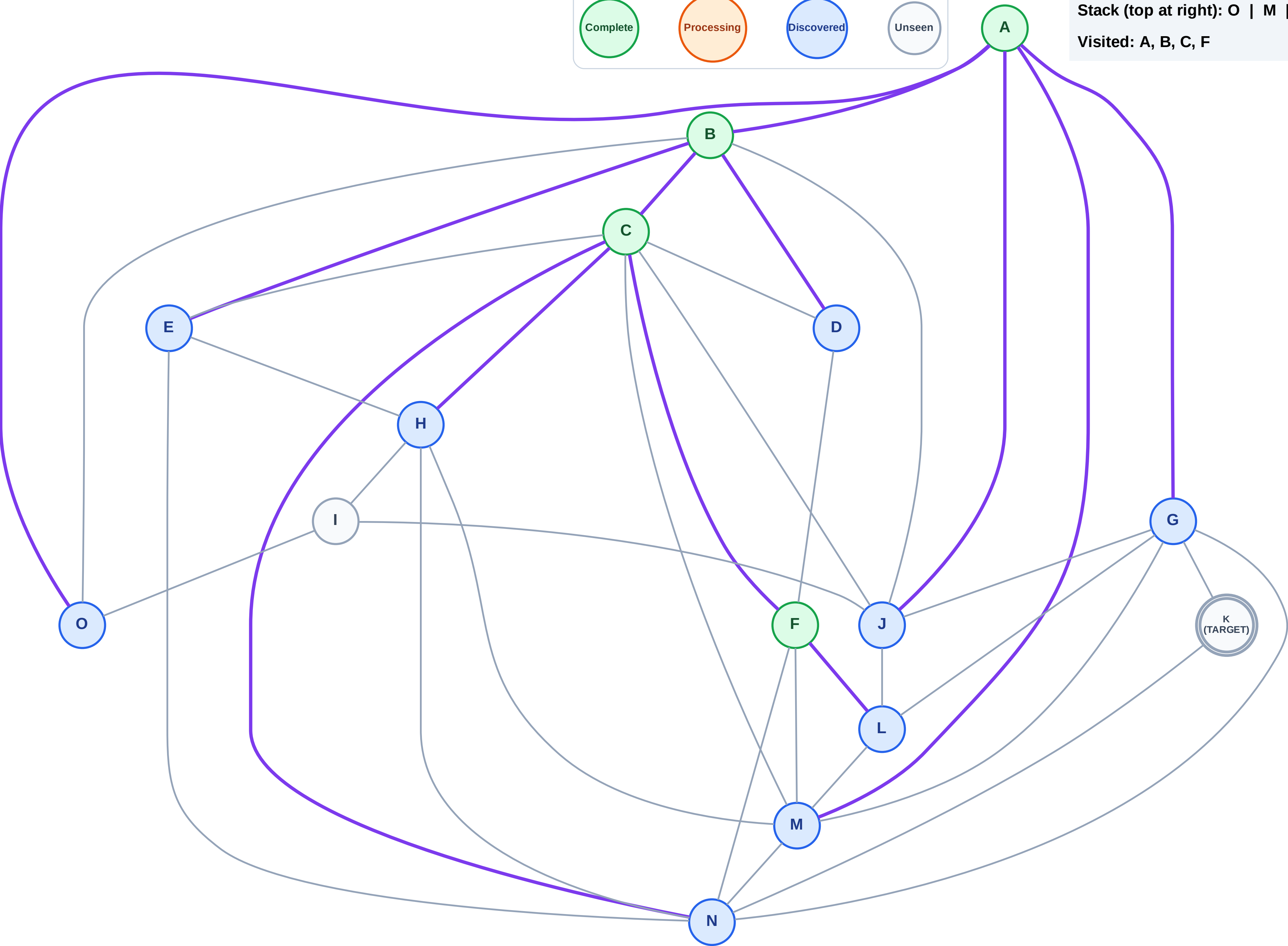
Processing

Discovered

Unseen

Stack (top at right): O | M | J | G | E | D | N | H | L

Visited: A, B, C, F



DFS Traversal

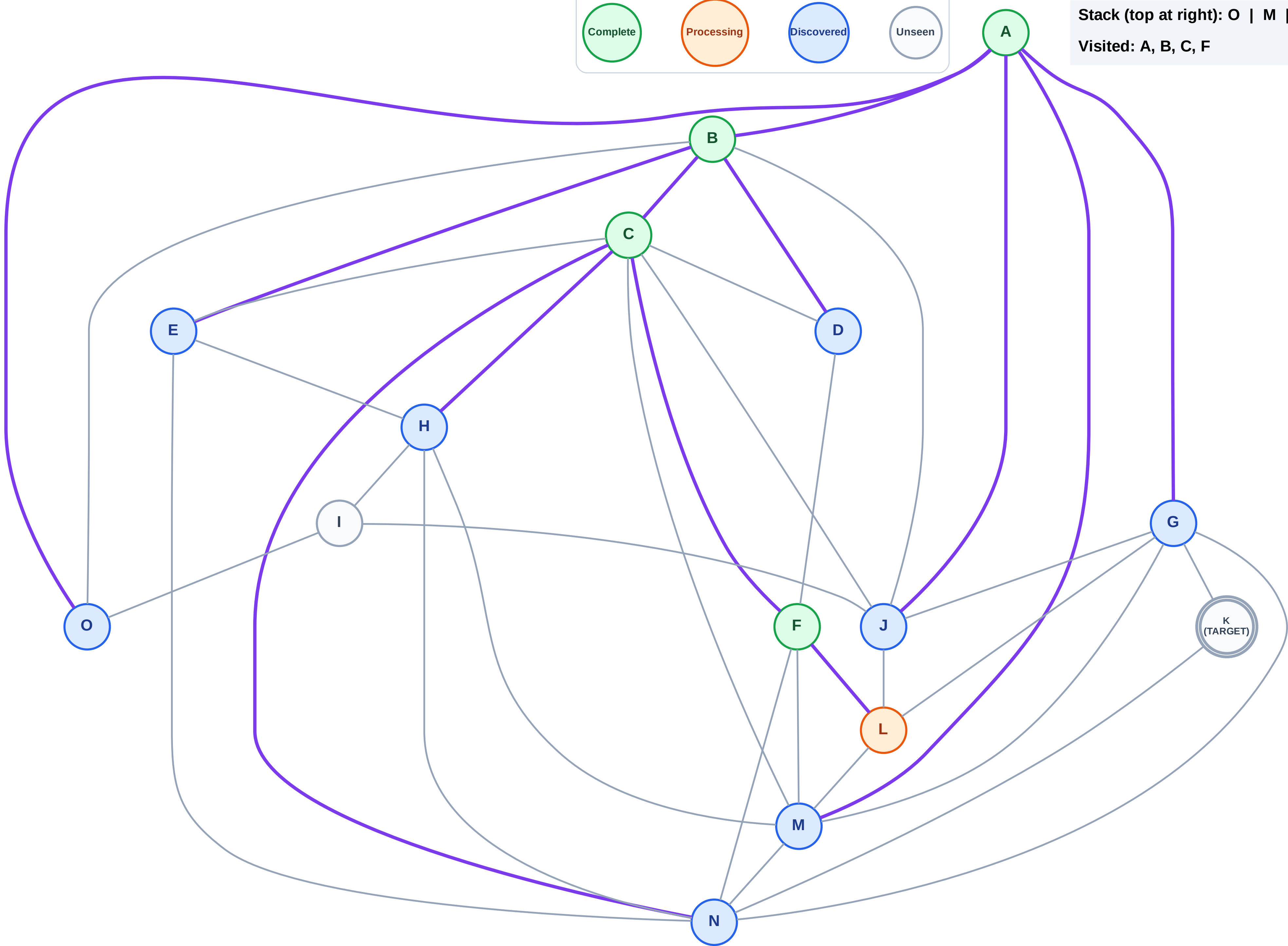
Step 10: Process node L

Legend



Stack (top at right): O | M | J | G | E | D | N | H

Visited: A, B, C, F



DFS Traversal

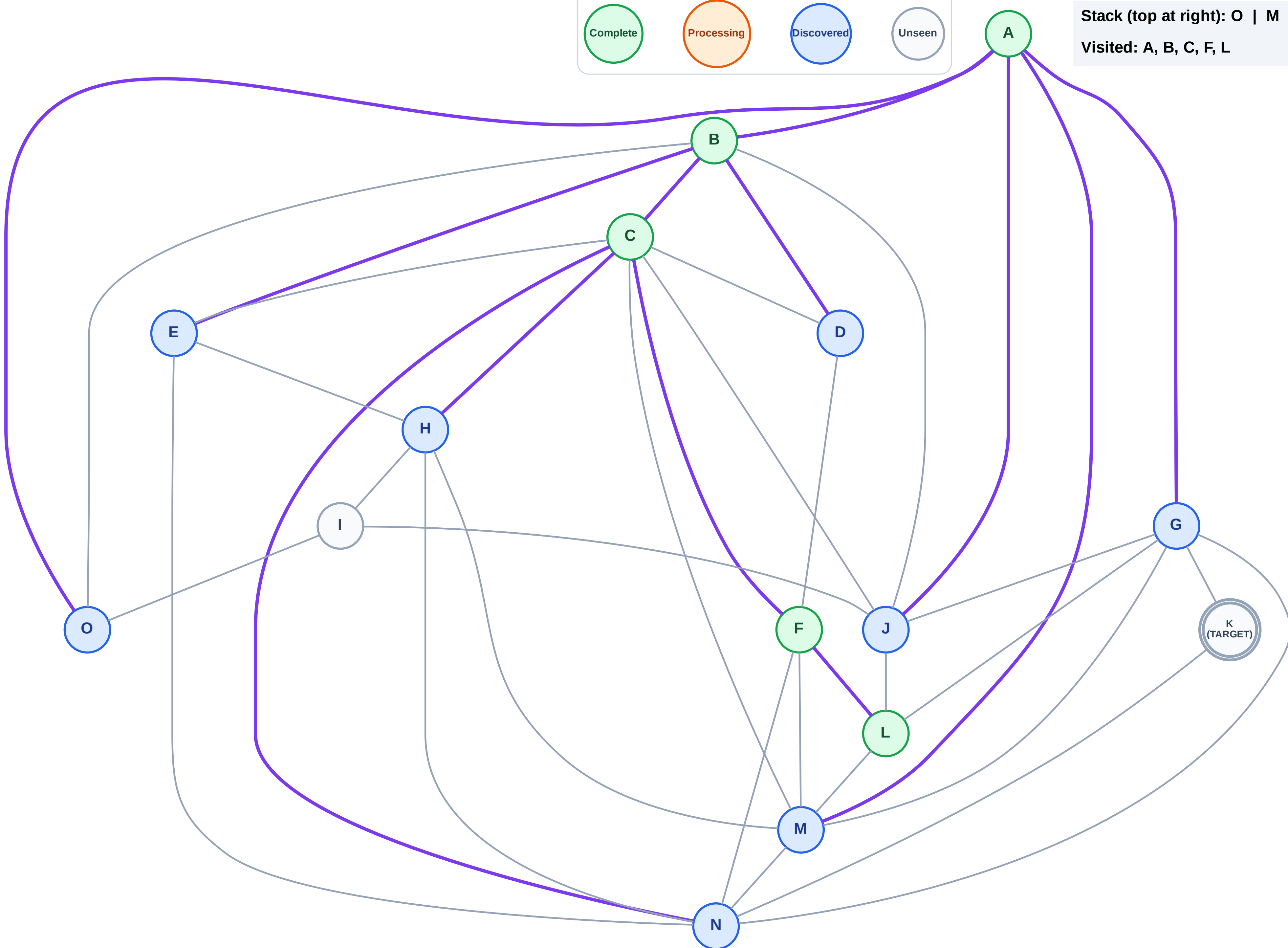
Step 11: No new neighbors from L

Legend



Stack (top at right): O | M | J | G | E | D | N | H

Visited: A, B, C, F, L



Step 12: Process node H

Legend



DFS Traversal

Step 13: Discover I from H

Legend

Complete

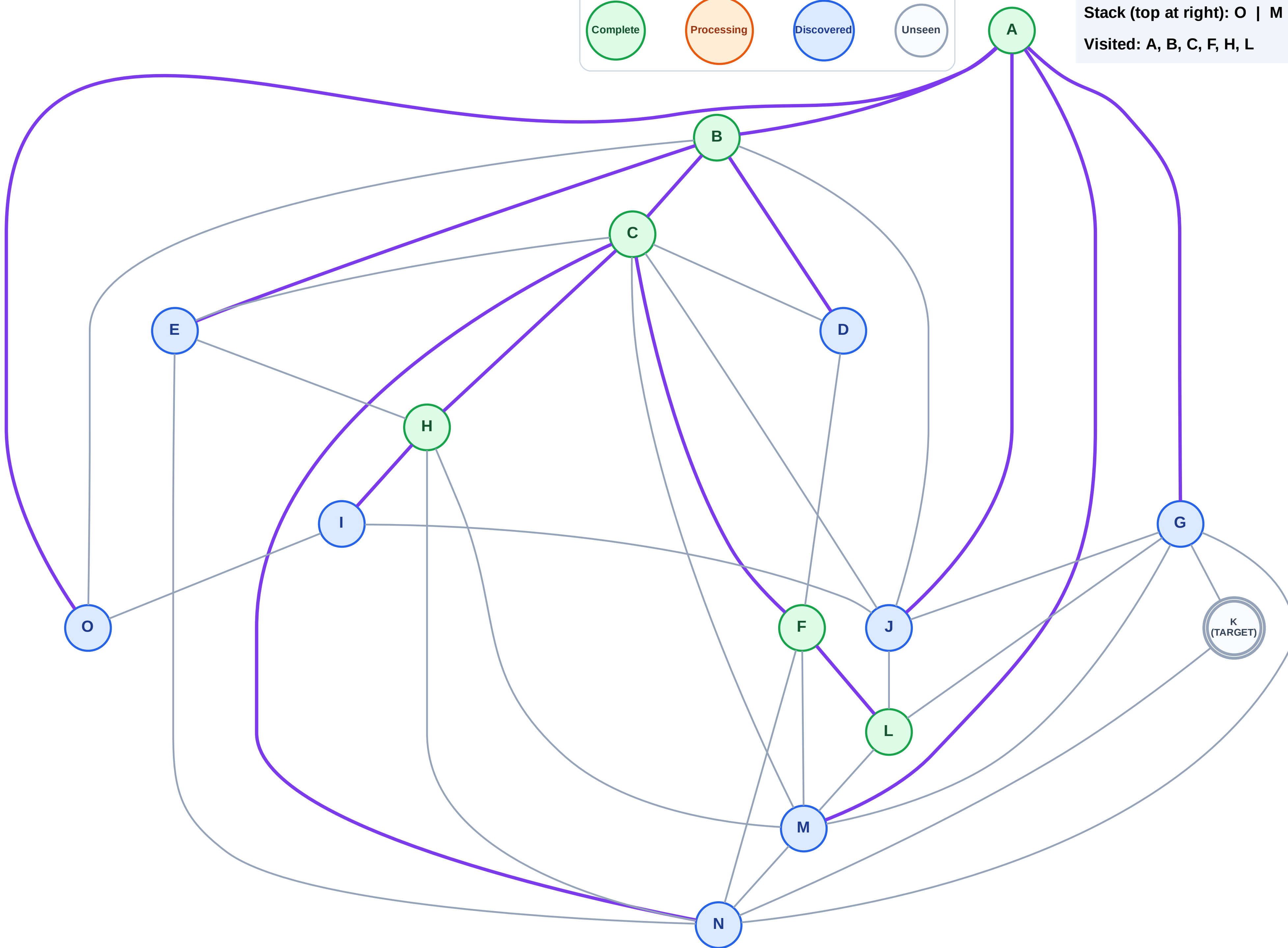
Processing

Discovered

Unseen

Stack (top at right): O | M | J | G | E | D | N | I

Visited: A, B, C, F, H, L



DFS Traversal

Step 14: Process node I

Legend

Complete

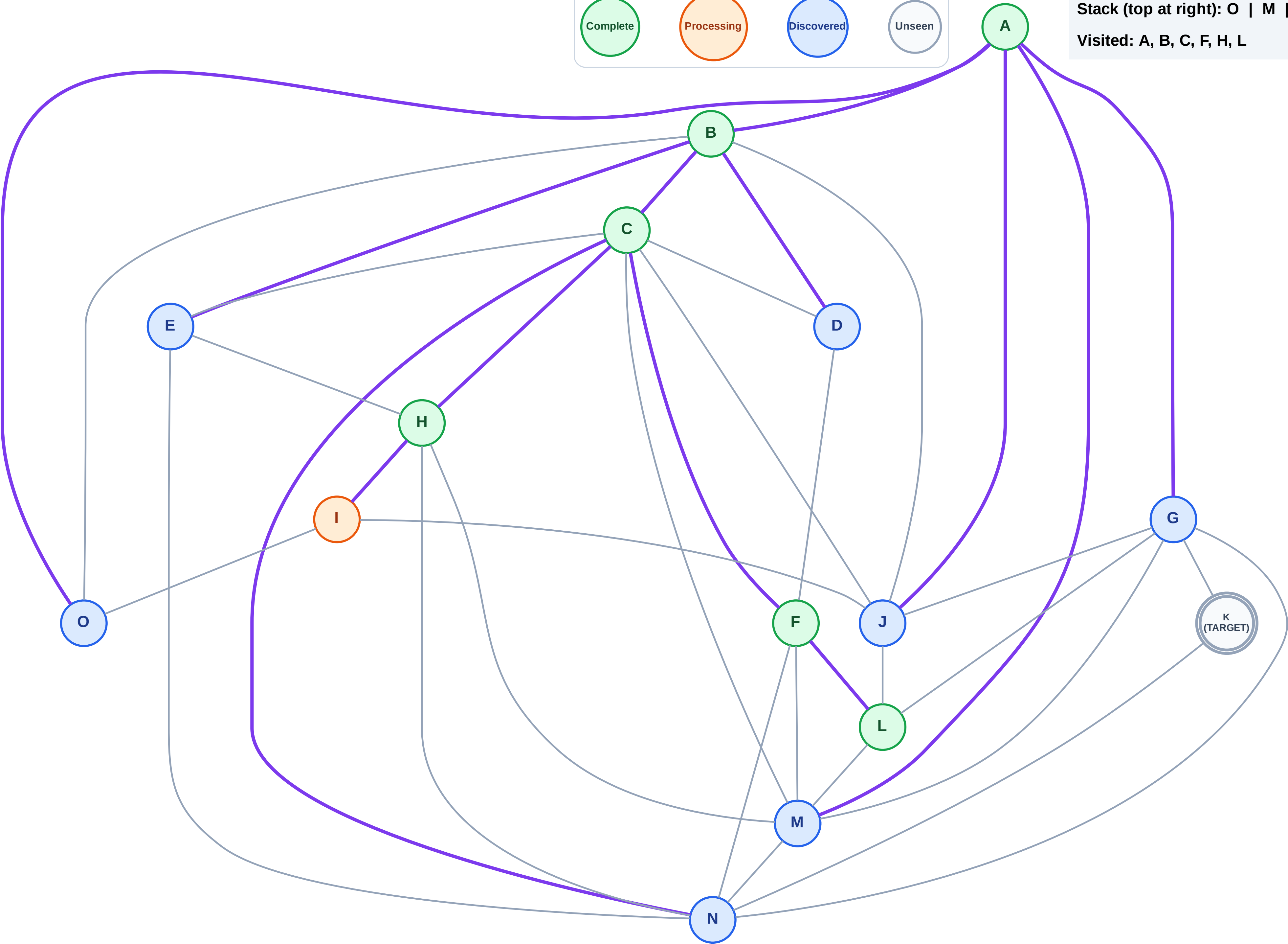
Processing

Discovered

Unseen

Stack (top at right): O | M | J | G | E | D | N

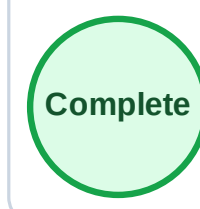
Visited: A, B, C, F, H, L



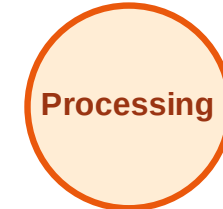
DFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



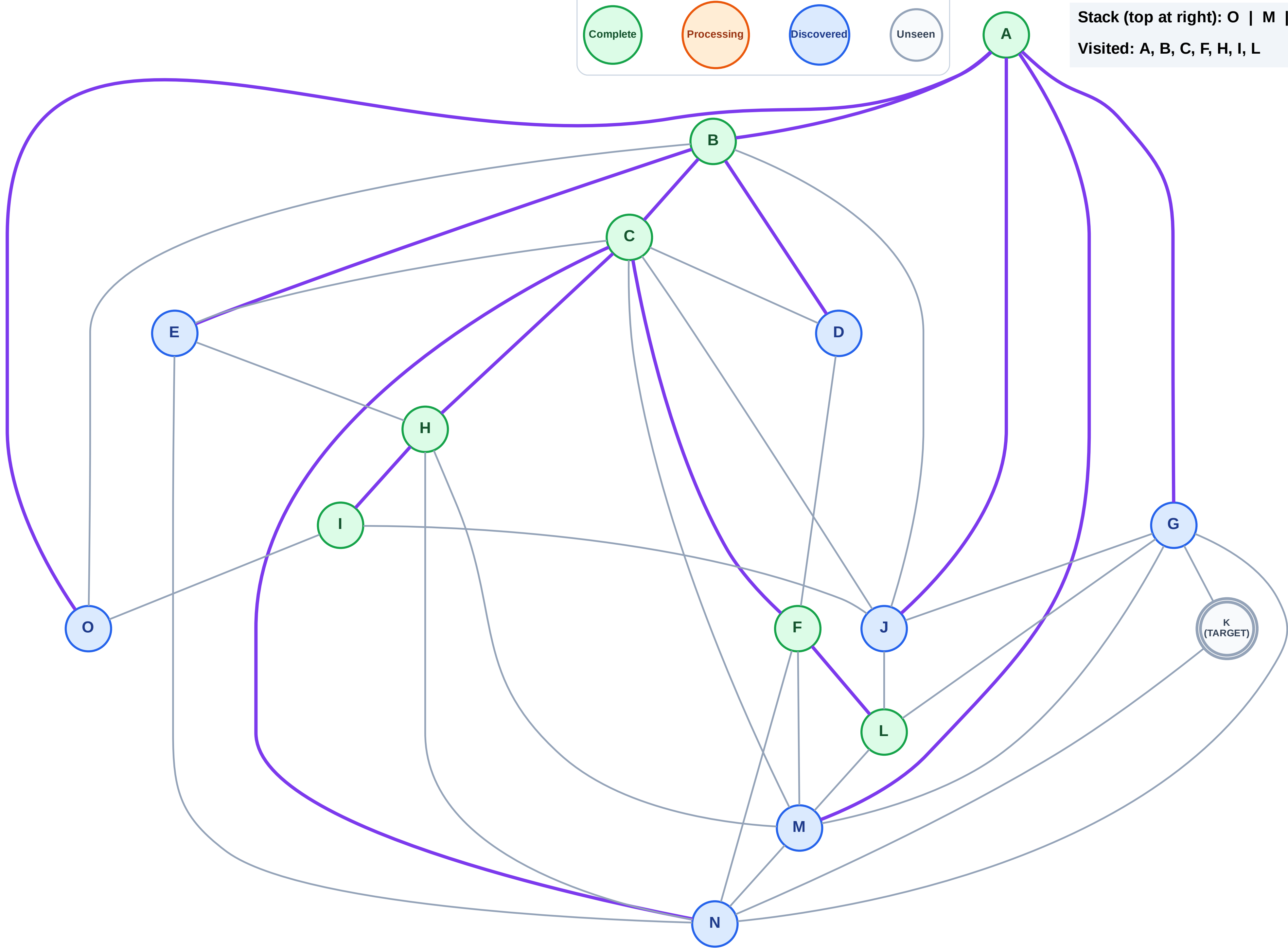
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | N

Visited: A, B, C, F, H, I, L



DFS Traversal

Step 16: Process node N

Legend



Complete



Processing

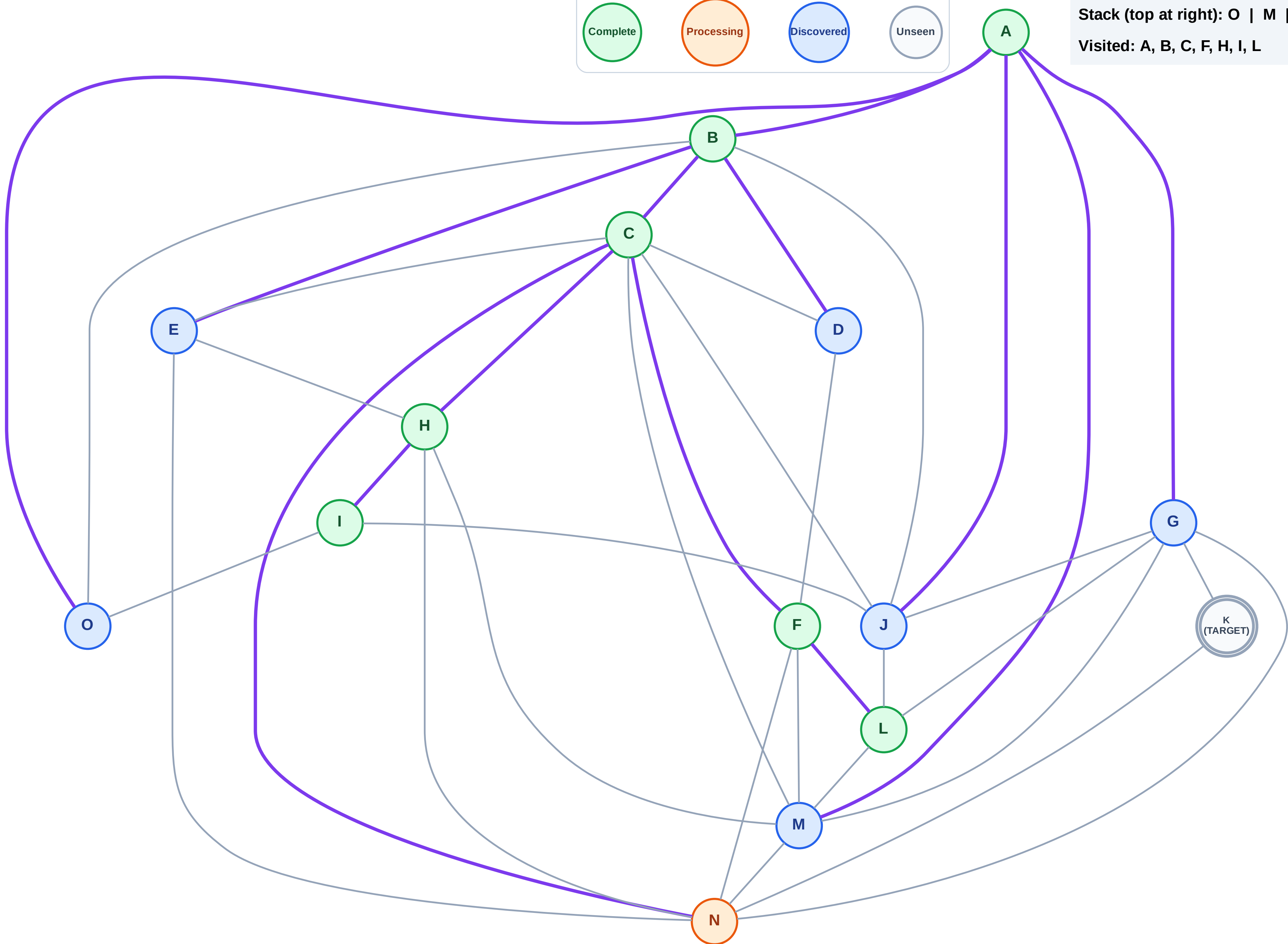


Discovered



Unseen

Stack (top at right): O | M | J | G | E | D
Visited: A, B, C, F, H, I, L



DFS Traversal

Step 17: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → C → N → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | J | G | E | D

Visited: A, B, C, F, H, I, K, L, N

